Group Theory

Algebric Structures

- **Binary operation**: Let G be a non-empty set. Any function or mapping from $G \times G$ to G itself is called binary operation on a set G i.e. a function $f: G \times G \to G$ is a binary operation on a set G.

  An operation is said to be Binary if it satisfies closure property.

- **Algebraic structure**: A non-empty set equipped with one or more binary operations is called algebraic structure. The algebraic structure consisting of a set G and binary operation $*, \circ$ on G is denoted by $(G, *, \circ)$.

  Group is a algebraic structure with one binary operation.

- **Closure property**: For every pair a and $b \in S$, $a * b$ is in S, then we say S is closed with respect to the binary operation "$*$".

  Such an algebraic structure $(S, *)$ is called groupoid.

- **Associative property**: If for any three elements $a, b, c \in S$

  $a * (b * c) = (a * b) * c$ holds.

  Such an algebraic structure satisfying closure & associative property is called semi-group.

- **Identity Property**: If there exists an element 'e' in S, such that $a * e = e * a = a$, for every $a \in S$, then a is called identity element.

"0" is called the Additive identity.

"1" is called the Multiplicative identity".

Such an algebraic structure satisfying closure, Associative and identity property is called Monoid.

- Inverse Property: For every a in S, if there exist an element a' such that

  $a * a' = a' * a = e$ then "a'" called inverse of "a".

"-a" is the additive inverse of "a".

"1/a" is the Multiplicative inverse of "a".

Such an algebraic structure satisfying closure, Associative, Identity and Inverse property is called Group.

- ~~Abelian~~ Commutative Property: If for all a and b in a group $a * b = b * a$, then the group is called commutative group or Abelian group.

For ex. $(Z, +)$ is an Abelian group.

- order of a group: If in a group $(G, *)$, the underlying set G consists of a finite no. of distinct elements, then the group is called finite group or otherwise infinite group.

The no. of elements in a finite group is called the order of the grour and is denoted by $O(G)$ or $|G|$.

St. the set D of all odd integers forms a commutative group with respect to the composition * defined by $a * b = a+b-1$, for all $a, b \in D$.

$D = \{\ldots -7, -5, -3, -1, 1, 3, 5, 7, \ldots\}$

1. closure property:

$\forall\ a, b \in D$

$a * b = a+b-1$ is always an odd integer

$\therefore a * b \in D$

$\therefore$ 'D' is a closure w.r.t '*'.

2. Associative property:

Let $a, b, c \in D$ be any 3 odd integers.

Now, $a * (b * c) = a * (b+c-1)$

$= a+b+c-1-1$

$= a+b+c-2$

$(a * b) * c = (a+b-1) * c$

$= a+b-1+c-1$

$= a+b+c-2$

$\therefore$ '*' is satisfying the associative property on 'D'.

3. Identity Property:

Let $a \in D$ be any odd integer

Now, let e be the identity element.

$\therefore a * e = e * a = a$

i.e. $a * e = a$

$a+e-1 = a$

$e = 1 \in D$

$\therefore$ '*' is satisfying the identity property on 'D'.

4. Inverse property :

Let a be any odd integer $\in D$.

Now let a' be the inverse of 'a'.

$$a * a' = a' * a = e$$

$$a * a' = e$$

$$a + a' - 1 = e$$

$$a + a' - 1 = 1$$

$$a + a' = 2$$

$$a' = 2 - a \in D$$

$\therefore$ '*' satisfying inverse property on D.

$\therefore$ (D, *) is a group.

Now.

5. Commutative property :

Let $a, b \in D$ be any odd integer.

Now, $a * b = a + b - 1$

$b * a = a + b - 1$

$\therefore a * b = b * a$.

$\therefore$ '*' is commutative on 'D'.

Hence, (D, *) is a commutative group.

Show that, the set of all roots of the equation $x^4 = 1$ forms a commutative group under multiplication.

$x^4 = 1$

$x^4 - 1 = 0$

$(x^2)^2 - 1 = 0$

$(x^2 + 1)(x^2 - 1) = 0$

$x^2 = -1 \qquad x^2 = +1$

$x = \pm\sqrt{-1} \qquad x = \pm\sqrt{1}$

$= \pm i \qquad = \pm 1$

$x = +i, -i, 1, -1$

$\therefore S = \{1, -1, i, -i\}$

$\cdot$	1	-1	i	$-i$
1	1	-1	i	$-i$
-1	-1	$+1$	$-i$	i
i	i	$-i$	i^2 (-1)	1
$-i$	$-i$	i	1	-1

(i) **closure**: From the above table it is clear that product of any two elements of S belongs to S.

$\therefore$ S is closed w.r.t $\cdot$.

(ii) **Associative**: Since multiplication is associative on the set $\mathbb{C}$, so multiplication is associative so S as well.

(3) Identity : Here 1 is the multiplicative identity and $1 \in S$.

So. multiplication satisfies identity property on S.

(4) Inverse : Inverse elements of all the elements of S belongs to S.

∴ Multiplication is satisfying on S.

Inverse property.

(5) commutative : Since the composition table is symmetric w.r.t the principle diagonal

∴ $(S, \cdot)$ is a commutative group.

■ Order of an element : Let $(G, *)$ be a group and a belongs to G. Then the order of "a" is the least positive integer 'n' such that

$$a^n = e$$

The identity element of G and is denoted by $O(a)$. If there exists no positive integer 'n' such that $a^n = e$, then the order of a is said to be infinity or zero.

For ex. $(S, \cdot)$ where $S = \{1, -1, i, -i\}$

Identity element is '1'

order of "-1" i.e. $O(-1) = 2$ ∵ $(-1)^2 = 1$

order of 'i' = $O(i) = 4$

order of '-i' = $O(-i) = 4$

If b is an element of a group and order of b is 20. find order of b^{15}.

$o(b) = 20$

$o(b^{15}) = ?$

Let $a = b^{15}$

Since, $o(b) = 20$, hence $b^n = e$ iff $20 | n$

So, $a = b^{15} \neq e$, as $20 \nmid 15$

$a^2 = (b^{15})^2 = b^{30} \neq e$ as $20 \nmid 30$

$a^3 = (b^{15})^3 = b^{45} \neq e$ as $20 \nmid 45$

$a^4 = (b^{15})^4 = b^{60} = e$ as $20 | 60$

$\therefore o(b^{15}) = 4$.

Let G be a group and 'a' belongs to G. If $o(a) = 24$. Find $o(a^4)$.

$o(a) = 24$ $o(a^4) = ?$

Let $b = a^4$

Since, $o(a) = 24$ hence $a^n = e$ iff $24 | n$

So. $b = a^4 \neq e$, as $24 \nmid 4$

$b^2 = a^8 \neq e$, as $24 \nmid 8$

$b^3 = a^{12} \neq e$, as $24 | 12$

$b^4 = a^{16} \neq e$ as $24 \nmid 16$

$b^5 = a^{20} \neq e$ as $24 \nmid 20$ $\therefore$

$b^6 = a^{24} = e$ as $24 | 24$.

$\therefore o(a^4) = 6$

If G be the group such that $(ab)^2 = a^2b^2$ for $a, a.b \in G$, show that the group G is Abelian.

or.

Prove that, a group $(G, *)$ is commutative iff $(a*b)^2 = a^2 * b^2$, for all $a, b \in G$.

Let us first assume that $(G, *)$ is Abelian.

$\therefore a * b = b * a, \forall a, b \in G$

$(a*b)^2 = (a*b) * (a*b)$

$= a * (b * a) * b$ [Associative]

$= a * (a * b) * b$ [$\because (G, *)$ is Abelian]

$= (a * a) * (b * b)$ [Associative]

$= a^2 * b^2$ [Proved]

Now, let us assume $(a*b)^2 = a^2 * b^2$

$(a*b) * (a*b) = (a*a) * (b*b)$

$a * (b * a) * b = a * (a * b) * b$

[Associative]

$\therefore b * a = a * b$

[by using left & right cancellation]

$\therefore (G, *)$ is Abelian.

Let $(G, *)$ be a group and $a, b \in G$. Suppose that $a^2 = e$ and $a * b^4 * a = b^7$. Prove that $b^{33} = e$.

$$a * b^4 * a = b^7$$

$$\Rightarrow (a * b^4 * a) * (a * b^4 * a) = (b^7)^2$$

$$\Rightarrow a * b^4 * (a * a) * b^4 * a = b^{14} \quad \text{[Associative]}$$

$$\Rightarrow a * b^4 * e * b^4 * a = b^{14}$$

$$\Rightarrow a * (b^4 * b^4) * a = b^{14}$$

$$\Rightarrow a * b^8 * a = b^{14}$$

$$b^{14} = a * b^8 * a$$

$$= a * b^7 * b * a$$

$$= a * (a * b^4 * a) * b * a$$

$$= (a * a) * b^4 * (a * b * a)$$

[Associative]

$$b^{14} = b^4 * a * b * a$$

$$b^4 * b^{10} = b^4 * a * b * a$$

$$b^{10} = a * b * a \quad \text{[Left cancellation law]}$$

$$(b^{10})^4 = (a * b * a) \quad \text{[raising to the power 4]}$$

$$b^{40} = (a * b * a) * (a * b * a) * (a * b * a) * (a * b * a)$$

$$= a * b * (a * a) * b * (a * a) * b * (a * a) * b * a$$

$$b^{40} = a * b^4 * a \quad \text{--- (i)}$$

$$b^7 = a * b^4 * a \quad \text{--- (ii)}$$

comparing (i) and (ii).

$$b^{40} = b^7$$

$$b^{40-7} = e$$

$$b^{33} = e \quad \text{[Proved]}$$

Subgroup : A non-empty subset H of a group (G, o) is said to be a subgroup of G, if H itself is a group under the binary operation 'o' of G.

Since every set is a subset of itself, so G itself is a subgroup of G. Again the subgroup of G containing only identity element is also a subgroup of G. Thus every group (G, o) has atleast two subgroups. These two are called trival or improper subgroups. All other subgroups are called non-trivial or proper subgroups.

Theorem 1 : The identity element of a subgroup is the same as that of the group.

Theorem 2 : The inverse of any element of a subgroup is the same as the inverse of the same regarded as an element in the group.

Theorem 3 : A non-empty subset H of a group (G, o) is a subgroup of G if and only if —

(i) $a, b \in H \Rightarrow a \circ b \in H$

(ii) $a \in H \Rightarrow a^{-1} \in H$.

Theorem 4 : The necessary and sufficient condition for a non-empty subset H of a group (G, o) to be a subgroup is that $a, b \in H \Rightarrow a \circ b^{-1} \in H$.

Theorem 5 : Intersection of any two subgroups of a group (G, o) is a subgroup of G.

Proof:

Let H_1 and H_2 be two subgroups of G.

Let $a, b \in H_1 \cap H_2 = H$ be any two elements

$\therefore a, b \in H_1$ and $a, b \in H_2$

$\therefore a \circ b^{-1} \in H_1$ and $a \circ b^{-1} \in H_2$

$a \circ b^{-1} \in H_1 \cap H_2$

$\therefore H_1 \cap H_2$ is a subgroup of G (Proved).

Note: The union of any two subgroups of a group is not necessarily a subgroup.

Let $(Z, +)$ be the group w.r.t '+'

$H_1 = \{0, \pm 2, \pm 4, \pm 6, \pm 8 \ldots\}$

$H_2 = \{0, \pm 3, \pm 6, \pm 9, \ldots\}$

$H_1 \cup H_2 = \{0, \pm 2, \pm 3, \pm 4, \pm 6, \pm 9 \ldots\}$

$2 + 3 = 5 \notin H_1 \cup H_2$

$\therefore H_1 \cup H_2$ is not a subgroup of $(Z, +)$.

Theorem 6: The union of any two subgroup of a group $(G, \circ)$ is a subgroup of G if and only if one is contained in the other.

- Product of two subgroups:

If H and K are two subgroups of the group $(G, \circ)$ then

$HK = \{x : x = h \circ k, h \in H, k \in K\}$

If H and K are two subgroups of a group G. Then prove that HK is also subgroup of G. iff $HK = KH$.

Let HK is a subgroup of G.

Let $x \in HK \Rightarrow x^{-1} \in H$

Now let $x^{-1} = h_1 k_1$ where $h_1 \in H$, $k_1 \in K$

$\therefore x = (x^{-1})^{-1}$

$= (h_1 k_1)^{-1}$

$= k_1^{-1} h_1^{-1} \in KH$

$\therefore HK \subset KH$. — (i)

Let $k_2 h_2 \in KH$ be any element.

$\therefore k_2 \in K$, $h_2 \in H$

$k_2^{-1} \in K$ & $h_2^{-1} \in H$ ($\because H$ & K are subgroups]

$h_2^{-1} k_2^{-1} \in HK$

$(h_2^{-1} k_2^{-1})^{-1} \in HK$

$k_2^{**} h_2^{**} \in HK$

$\therefore KH \subset HK$. — (ii)

$\therefore HK = KH$

Now, conversely let, $HK = KH$

Let $p, q \in HK$ be any two elements

$p = h_3 k_3$ & $q = h_4 k_4$ where $h_3, h_4 \in H$

$k_3, k_4 \in K$

$pq = (h_3k_3)(h_4k_4)$

$= h_3(k_3h_4)k_4$ [Associative]

$= h_3(h_4k_3).k_4$ [$\because HK \subset KH, \therefore h_4k_3 = k_3h_4$]

$= (h_3h_4)(k_3k_4)$ [associative]

$= h_5k_5 \in HK$ — (iii)

$\therefore pq \in HK \Rightarrow pq \in HK$

$p = h_3k_3 \quad \therefore p^{-1} = (h_3k_3)^{-1}$

$= k_3^{-1}h_3^{-1} \in KH = HK$

$\therefore p^{-1} \in HK$ — (iv)

from (iii) & (iv),

HK is a subgroup of G.

Modulo Groups :

Residue classes modulo m :

As the congruence modulo m is an equivalence relation in Z. so, it forms a partition Z into m disjoint equivalence classes which are called residue classes modulo m, or congruence classes modulo m. They are denoted by $[0], [1], [2], \ldots [m-1]$

we denote the set of all residue classes modulo m in Z by Z_m.

Thus $Z_m = \{[0], [1], \ldots [m-1]\}$

The residue class $[0]$ is called zero residue class.

For example $Z_4 = \{[0], [1], [2], [3]\}$:

where,

$[0] = \{0, \pm 4, \pm 8, \ldots\}$

$[1] = \{1, 1 \pm 4, 1 \pm 8, \ldots\}$

$[2] = \{2, 2 \pm 4, 2 \pm 8, \ldots\}$

$[3] = \{3, 3 \pm 4, 3 \pm 8, \ldots\}$

Show that, the set of residue classes modulo 4 forms a group of order 4 w.r.t addition of residue classes.

$Z_4 = \{[0], [1], [2], [3]\}$

Composition table:

$\oplus$	$[0]$	$[1]$	$[2]$	$[3]$
$[0]$	$[0]$	$[1]$	$[2]$	$[3]$
$[1]$	$[1]$	$[2]$	$[3]$	$[0]$
$[2]$	$[2]$	$[3]$	$[0]$	$[1]$
$[3]$	$[3]$	$[0]$	$[1]$	$[2]$

(i) Closure property:

Since, all the entries of are elements of Z_4 so, Z_4 is closed w.r.t $\oplus$.

(ii) Associative:

$[1], [2], [3]$

$([1] \oplus [2]) \oplus [3] = [1] \oplus ([2] \oplus [3])$

$\bar{2} = \bar{2}$

$\therefore$ The operation '$\oplus$' is associative on Z_4.

(iii) Identity :

$[1] \oplus [0] = [1]$

$[0] \in Z_4$ is the identity element.

(iv) Inverse :

$[1] \oplus [3] = [0]$

Inverse of every residue class present in Z_4

$\therefore (Z_4, \oplus)$ is a group.

(v) commutative :

$[a] \oplus [b] = [b] \oplus [a] \quad \forall\ a, b \in Z_4$

$\therefore (Z_4, \oplus)$ is a commutative group.

COSET

Right coset & left coset:

Let H be a subgroup of a group G. Also let a be an element of G. Then the subset $\{ha : h \in H\}$ is called a right coset of H in G and is denoted by Ha.

Thus $Ha = \{ha : h \in H\}$

Similarly, $aH = \{ah : h \in H\}$ is called left coset of H in G.

Note: (i) Since $He = H = eH$, so H itself is a right as well as a left coset of H.

(ii) Again since H is a subgroup of G, so $e \in H$ and $ea = a \in H$. Thus if Ha is any right coset then atleast one element a Ha.

Consequently every right coset is non-empty.

Similarly every left coset is non-empty.

(iii) when G is an Abelian group, then $ah = ha$ $\forall h \in H$ and hence the right coset Ha is equal to the corresponding left coset aH. But if the group G is not Abelian, then we may have $aH = Ha$ or $aH \neq Ha$.

Illustration:

Consider the additive group of $(\mathbb{Z}, +)$

where $\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3 \ldots\}$

$H = 3\mathbb{Z} = \{0, \pm 3, \pm 6, \pm 9 \ldots\}$ be a subgroup of $(\mathbb{Z}, +)$.

-6	-5	-4
-3	-2	1
0	1	2
3	4	5
6	7	8
9	10	11
H	1+H	2+H

we get three distinct coset H, 1+H, 2+H

$\therefore Z = H \cup (1+H) \cup (2+H)$.

Theorem 1: If H is any subgroup of a group G and $h \in H$, then $hH = H = Hh$.

Theorem 2: Let H be a subgroup of a group G and let $a, b \in G$. Then

(i) $aH = bH$ iff $a^{-1}b \in H$.

(ii) $Ha = Hb$ iff $ba^{-1} \in H$.

Theorem 3: Any two right or left coset of a subgroup are either disjoint or identical.

Theorem 4: Let H be a subgroup of a group G. Then there is a one-to-one correspondance between any two right (left) coset of H in G.

Theorem 5: Let H be a subgroup of a group G. Then there is a one-to-one correspondance between the set of left and the set of right coset of H in G.

Index of a subgroup:

Let H be a subgroup of a group G. Then the no. of distinct right coset of H in G is called the left index of H in G and is denoted by $[G:H]$ or by $i_G(H)$.

Theorem 6 (Lagrange's Theorem):

The order of each subgroup of a finite group is a divisor of the order of the group.

Theorem 8: Let G be a group of finite order n and $a \in G$. Then $o(a)$ divides n and $a^n = e$.

Proof: Let H be a subgroup of a finite group G.

Also let, $o(G) = n$

$o(H) = m$

Let $H = \{h_1, h_2, \ldots, h_m\}$

then each coset Ha for $a \in G$ has m right distinct elements $h_1a, h_2a, h_3a, \ldots h_ma$.

Since, G is finite, the no of distinct right cosets of H in G is finite.

Let $Ha_1, Ha_2, \ldots Ha_k$ be k distinct right cosets of H in G.

$G = Ha_1 \cup Ha_2 \cup \ldots \cup Ha_k$

$o(G) = o(Ha_1) + o(Ha_2) + \ldots + o(Ha_k)$

$o(G) = m + m + \ldots + mk$

$k = n/m$

$\Rightarrow o(H)$ is a divisor of $o(G)$.

Cyclic groups :

A group (G, o) is called cyclic if there exist an element a in G such that every element of G can be expressed as a^n where n is an integer. The element a is called the generator of the cyclic group.

Thus if a be the generator of the cyclic Group G, then $G = \{a^n : n \in Z\}$ which is denoted by $G = \langle a \rangle$ or $\{a\}$.

Ex. Hour hand of clock:

12'o clock $= r^0$
1'o clock $= r^1$
2'o " $= r^2$
:
12'o clock $= r^0$.

$S = \{r^0, r^1, r^2, r^3 \ldots r^{11}\}$

$S = \{r^n : n = 0, 1, \ldots 11\}$

$S = \langle r \rangle$.

Ex. The multiplicative group $G = \{1, -1, i, -i\}$ is cyclic. as we can write $G = \{i, i^2, i^3, i^4\}$
Here i is a generator of G.

Example of infinite cyclic group : $(Z, +)$.

Theorem: Every cyclic group is an Abelian group.

Proof: Let, G be a cyclic group.

Let us take $a, b \in G$.

Since, G is cyclic, it is generated by some $g \in G$.

So, $a = g^i$ and $b = g^j$ for some $i, j \in \mathbb{Z}$.

Now, $ab = g^i g^j$

$= g^{i+j}$

$= g^{j+i}$

$= g^j \cdot g^i$

$= ba$.

$\therefore \forall a, b \in G$ we have $ab = ba$.

Hence, G is Abelian group.

Note: Converse is not true.

Give an example of an Abelian group which is not cyclic.

Klein's 4 group, which is the power set of $\{1, 2\}$ with the symmetric difference (Δ) as the binary operation.

$A = \{1, 2\}$ $\quad P(A) = \{\{1\}, \{2\}, \{1, 2\}, \phi\}$.

$K_4 = \{P(A), \Delta\}$

$\langle \{1\} \rangle = \{\{1\}, \phi\}$

$\langle \{2\} \rangle = \{\{2\}, \phi\}$ and so on.

So, no element has order 4, which is the order of the group.

$\therefore K_4$ is not cyclic.

Let $S = \{z \in \mathbb{C} : z^n = 1\}$ where $\mathbb{C}$ is the set of all complex numbers. Pt. S forms a cyclic group w.r.t usual multiplication on $\mathbb{C}$.

or,

St. the n-th roots of unity form a cyclic group under ordinary multiplication.

$z^n = 1$

or, $z^n = \cos 0 + i \sin 0$.

or, $z^n = \cos(0 + 2k\pi) + i \sin(0 + 2k\pi)$; $k = 0, \pm 1, \pm 2, \ldots$

or, $z^n = \cos(2k\pi) + i \sin(2k\pi)$

or, $z = [\cos(2k\pi) + i \sin(2k\pi)]^{1/n}$ [$k = 0, 1, 2, \ldots (n-1)$]

$= \cos\left(\frac{2k\pi}{n}\right) + i \sin\left(\frac{2k\pi}{n}\right)$

[De Moivre's theorem]

or, $z = e^{\frac{2k\pi}{n} i}$, $k = 0, 1, 2, \ldots n-1$

Thus the set S of nth roots of unity is given by,

$S = \left\{1, e^{\frac{2\pi}{n} i}, e^{\frac{4\pi}{n} i}, e^{\frac{6\pi}{n} i}, \ldots e^{\frac{2(n-1)\cdot\pi}{n} i}\right\}$

$= \{w^0, w, w^2, \ldots, w^{n-1}\}$ where, $w = e^{\frac{2\pi}{n} i}$

(i) Closure: $\because w^n = 1$

Let $w^i, w^j \in S$.

Now, $w^i w^j = w^{i+j} = w^r \in S$

where, $r \in \{0, 1, \ldots n\}$ is the remainder of $i+j$ upon division by n.

(ii) Associative :

since, $\mathbb{C}$ is associative wrt multiplication.

$\therefore$ S which is subset of $\mathbb{C}$ will also be associative.

(iii) Identity : we know that, 1 is the identity for complex multiplication.

Now. $1 = w^0 \in S$.

(iv). Inverse : For any $w^i \in S$. its inverse is $w^{n-i} \in S$.

since, $w^i \cdot w^{n-i} = w^n = 1$, so S is a group. w.r.t complex multiplication.

since, all the elements of S can be expressed as power of 'w'.

Hence S is a cyclic group generated by 'w'.

The group $(\mathbb{Q}, +)$ of all rational no's wrt usual addition is cyclic group — False.

Let if possible $\mathbb{Q}$ is cyclic.

$q \in \mathbb{Q}$ be a generator.

$$\mathbb{Q} = \{nq : n \in \mathbb{Z}\}$$

clearly, $q \neq 0$ since otherwise $\mathbb{Q} = \{0\}$.

Now. $q/2$ is rational no. and so must have $q/2 = nq$, which is not possible. as $n = 1/2 \notin \mathbb{Z}$

so, not cyclic.

Theorem : Let (G, o) be a finite cyclic group generated by a. Then $o(G) = n$ if and only if $o(a) = n$.

Theorem : Let $G = \langle a \rangle$ be a cyclic group of order n. Then for any integer r where $1 \leq r < n$, a^r is a generator of G if and only if r and n are relative primes.

or, Let G be a cyclic group of order n. Let g be a generator. Then g^i is a generator if and only if $\gcd(n, i) = 1$.

Theorem : Every group of prime order is cyclic.

Proof : Let G be a group of prime order 'p'.
Since $p \geq 2$, G has atleast one element other than the identity element e.

Let, $a \in G$ & $a \neq e$

we know that, the order of an element must divide the order of the group. so, $o(a) \mid p$.

Since p is prime that means either $o(a) = 1$ or $o(a) = p$.

But, $o(a) \neq 1$

since $a \neq e$ $\therefore o(a) = p$.

since $o(a) = o(G)$.

$\therefore G$ is a cycle group.

Generators:

A group $(G, \circ)$ is cyclic if there exist an element a in G such that every element of G can be expressed as a^n where n is an integ

Theorem: If a is a generator of a cyclic group G, then a^{-1} is also a generator of G.

Proof.

Let G be a cyclic group with generator a.

Since, a is a generator of G;

$$G = \{a^n : n \in \mathbb{Z}\} = \{(a^{-1})^{(-n)} : n \in \mathbb{Z}\}$$
$$= \{(a^{-1})^m : m \in \mathbb{Z}\}$$

writing $m = -n$, using the fact $-\mathbb{Z} = \mathbb{Z}$.

Hence, G is generated by a^{-1}.

Theorem: Every proper subgroup of an infinite cyclic group is infinite.

Theorem: Every subgroup of a cyclic group is cyclic.

Pl. every proper subgroup of a group of order 6, is cyclic.

Let, G be a group of order 6. & H be a proper subgroup of G.

Then, $o(H)$ is a divisor of 6. [by Lagrange's theorem]

But the divisors of 6 are 1, 2, 3 & 6

Since, H is a proper subgroup of G, then $o(H) < 6$.

Now, if $O(H) = 1 \Rightarrow H = \{e\}$ which is cyclic.

if $O(H) = 2 \rightarrow H$ is also cyclic. prime ordered group is also cyclic.

if $O(H) = 3 \rightarrow H$ is also cyclic.

Thus, every proper subgroup of order 6 is cyclic.

Normal Subgroups :

A subgroup H of a group G is said to be normal if $aH = Ha \ \forall \ a \in G$. Then in symbol, we write $H \triangleleft G$.

Note :

(1) $aH = Ha$ does not imply that $ah = ha \ \forall \ h \in H$. But, from the definition, it follows that every subgroup of a commutative group is a normal subgroup.

(2) Every ~~subgroup~~ group G has at least two normal subgroups, namely, G itself and $\{e\}$. These are called improper or trivial normal subgroups. All others are called proper normal subgroups.

(3) For normal subgroups left cosets & right cosets are same.

Simple group : A group having no proper subgroups is called a simple group.

Theorem : Let H be a subgroup of a group G. Then H is normal in G if and only if $ghg^{-1} \in H$ and $\forall \ g \in G$. $h \in H$.

Proof :

Let H be a subgroup of G.

Then, $xH = Hx \quad \forall x \in G$

Let $h_1 \in H$ then $xh_1 = h_2 x$ for some $h_2 \in H$.

$xh_1 = h_2 x$

$\Rightarrow x h_1 x^{-1} = h_2 \in H, \forall x \in G$

$\therefore xhx^{-1} \in H, \forall x \in G.$

Now let us assume that, $xhx^{-1} \in H, \forall x \in G$
$h \in H.$

Let $a \in Hx$

$\Rightarrow a = h_3 x, \quad h_3 \in H$

$\Rightarrow a = (xx^{-1}) h_3 x$

$= x(x^{-1} h_3 x)$ by associative property.

$= x\left[x^{-1} h_3 (x^{-1})^{-1}\right]$

$= xh_4 \in xH$

$\therefore Hx \subset xH.$

Similarly, $b \in xH.$

$b = xh_5 \quad h_5 \in H$

$b = xh_5 = x h_5 (x^{-1}x)$

$= (x h_5 x^{-1}) x$

$= \left[(x^{-1})^{-1} h_5 x^{-1}\right] x.$

$= h_6 x \in Hx$

$\therefore xH \subset Hx$

$\therefore xH = Hx.$

Theorem: A subgroup H of a group G is normal if and only if $xHx^{-1} = H \ \forall x \in G$.

Theorem: A subgroup H of a group G is normal subgroup of G if and only if each left cosets of H in G is a right coset of H in G.

Prove that. Every subgroup of a cyclic group is normal.

Let G be a cyclic group and H be a subgroup of G.

Let $a \in G$ and $h \in H$

Then, $aha^{-1} = aa^{-1}h$ [Since G is cyclic; G is Abelian]

$\Rightarrow aha^{-1} = eh$

$\Rightarrow aha^{-1} = h \in H$.

If G is a group and H is a subgroup of index 2 in G, Pt H is a normal subgroup of G.

Since, the index of H in G is 2. so there are only two distinct left or right cosets of H in G.

The 2 distinct left cosets are H, $G-H$. Also the 2 distinct right cosets are H & $G-H$.

Case I Let. $x \in H$

$xH = H$ — (i) & $Hx = H$ — (ii)

From (i) & (ii), $xH = Hx$.

Case II. Now, let $x \notin H$. $\Rightarrow x \in G-H$

$\therefore xH = G - H$ (as $G-H$ is only left coset)

$\therefore Hx = G - H$ [as $G-H$ is only right coset]

$\therefore xH = Hx \quad \forall x \in H$

$\therefore H$ is a normal subgroup of G.

St. the center of a group G given by $Z(G) = \{a \in G : a*g = g*a \text{ for all } g \in G\}$ is a normal subgroup of G.

Part 1: $Z(G)$ is a subgroup of G.

Part 2: Now for any $g \in G$ & any $a \in Z(G)$

we have $gag^{-1} = agg^{-1}$ [$\because ag = ga \ \forall g \in G$]

or $gag^{-1} = a \in Z(G)$

$\therefore g \in G, a \in Z(G)$

$\Rightarrow gag^{-1} \in Z(G)$

$\therefore Z(G)$ is a normal subgroup of G.

Pf. $H = \left\{ \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} \mid x \in R, x \neq 0 \right\}$ forms a subgroup of $GL(2,R)$, the group of all real non-singular 2×2 matrices.

Part 1: Let $A = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \in H$ & $B = \begin{pmatrix} b & 0 \\ 0 & b \end{pmatrix} \in H$

$\therefore a, b \neq 0$ & $a, b \in R$

Now, $\det(B) = \begin{vmatrix} b & 0 \\ 0 & b \end{vmatrix} = b^2 \neq 0 \quad \therefore B^{-1}$ exist

Now, $B^{-1} = \frac{adj\ B}{det(B)} = \frac{1}{b^2}\begin{pmatrix} b & 0 \\ 0 & b \end{pmatrix}$

$= \begin{pmatrix} 1/b & 0 \\ 0 & 1/b \end{pmatrix}$

$AB^{-1} = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}\begin{pmatrix} 1/b & 0 \\ 0 & 1/b \end{pmatrix}$

$= \begin{pmatrix} a/b & 0 \\ 0 & a/b \end{pmatrix} \in H$, as $a/b \in R$ and $a/b \neq 0$

Hence, $A, B \in H \Rightarrow AB^{-1} \in H$

$\therefore$ H is a subgroup.

Part II,

Let $C = \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix} \in G$

$\because$ C is non-singular. $|C| \neq 0$ so C^{-1} exist

$\therefore$ $det\,|C| = x_1y_2 - x_2y_1$

$C^{-1} = \frac{adj\,(C)}{det\,(C)} = \frac{1}{x_1y_2 - x_2y_1}\begin{pmatrix} y_2 & -y_1 \\ -x_2 & x_1 \end{pmatrix}$

$CIC^{-1} = \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix}\begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix}\frac{1}{x_1y_2 - x_2y_1}\begin{pmatrix} y_2 & -y_1 \\ -x_2 & x_1 \end{pmatrix}$

$= \frac{1}{x_1y_2 - x_2y_1}\begin{pmatrix} xx_1y_2 - xy_1x_2 & 0 \\ 0 & -xy_1x_2 + xy_1y_2 \end{pmatrix}$

$$= \begin{pmatrix} x\,(x_1y_2 - y_1x_2)\cdot\dfrac{1}{(x_1y_2 - x_2y_1)} & 0 \\ 0 & x\cdot\dfrac{(x_1y_2 - y_1x_2)}{(x_1y_2 - x_2y_1)} \end{pmatrix}$$

$$= \begin{pmatrix} x & 0 \\ 0 & x \end{pmatrix} = D \in H.$$

$\therefore$ H is a normal subgroup of G.

Permutation Group

Let S be a finite set having n distinct elements. Then an one-one mapping of S onto itself is called a permutation of S of degree n.

Let $S = \{a_1, a_2, \ldots, a_n\}$ be a finite set having n distinct elements and $f: S \to S$ be a bijective mapping defined by

$$f(a_i) = b_i,\ i = 1, 2, 3, \ldots, n.$$

Then the permutation f can be written as -

$$f = \begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ f(a_1) & f(a_2) & \cdots & f(a_n) \end{pmatrix} \text{ or } \begin{pmatrix} a_1 & a_2 & \cdots & a_n \\ b_1 & b_2 & \cdots & b_n \end{pmatrix}$$

Total no. of distinct permutations of degree n is n!. The set of all distinct permutations of degree n is called the symmetric set of permutations of degree n and is denoted by P_n or S_n.

Composite or product of two matrices:

$$\gamma = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \quad f = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$\gamma \circ f = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \\ 3 & 2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

Inverse of a permutation :

$$\gamma = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$\gamma^{-1} = \begin{pmatrix} 3 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}.$$

Equality of two permutations :

Two permutations f_1 and f_2 degree are said to be equal if $f_1(a_i) = f_2(a) \; \forall \; a \in S$.

$$f_1 = \begin{pmatrix} a & b & c \\ c & a & b \end{pmatrix}$$

$$f_2 = \begin{pmatrix} b & a & c \\ a & c & b \end{pmatrix} = \begin{pmatrix} a & b & c \\ c & a & b \end{pmatrix}$$

$\therefore f_1 = f_2$.

Identity permutation :

$$e = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \text{ or } \begin{pmatrix} a & b & c \\ a & b & c \end{pmatrix}.$$

Let $A = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{pmatrix}$

be two permutations. Show that $AB \neq BA$.

$$AB = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 5 & 4 & 3 \end{pmatrix}.$$

$$BA = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 1 & 2 & 5 \end{pmatrix}$$

$\therefore AB \neq BA$

Find the inverse of the following permutations:

(i) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix}$

$= \begin{pmatrix} 3 & 4 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 1 & 2 \end{pmatrix}$

(ii) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 6 & 5 & 4 & 2 \end{pmatrix}$

$= \begin{pmatrix} 3 & 1 & 6 & 5 & 4 & 2 \\ 1 & 2 & 3 & 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 6 & 1 & 5 & 4 & 3 \end{pmatrix}$

Cyclic Permutations:

$\gamma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 3 & 9 & 7 & 2 & 4 & 6 & 1 & 5 & 8 \end{pmatrix}$

cyclic notation $= (1,3,7)(2,9,8,5,4)(6)$

$= (1,3,7)(2,9,8,5,4)$

1. There will be distinct elements in every cycle.
2. can be started from any element.

Now do the reverse.

$\alpha = (1\ 3\ 4)(5\ 6)(2\ 7\ 8\ 9)$

$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 3 & 7 & 4 & 1 & 6 & 5 & 8 & 9 & 2 \end{pmatrix}$

- **Transposition:** A cycle of length '2'.

- **Multiplication of cyclic permutation:**

$\pi = (1,2,3) \qquad \sigma = (5\ 4\ 1)$

$A = \{1, 2, 3, 4, 5\}$

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix} \qquad \sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 3 & 1 & 4 \end{pmatrix}$$

$$\pi\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 1 & 2 & 4 \end{pmatrix}.$$

$= (1, 5, 4, 2, 3)$

- **Order of a permutation:**

$\pi^n = e$

n is the least positive integer.

$o(\pi) = n.$

Ex. $\pi = (1\ 2)(3\ 4\ 5) \qquad A = \{1, 2, 3, 4, 5\}$

$\pi^0 = (1\ 2)(3\ 4\ 5)$

$\pi^1 = (2\ 1)(4\ 5\ 3)$

$\pi^2 = (1\ 2)(5\ 3\ 4)$

$\pi^3 = (2\ 1)(3\ 4\ 5)$

$\pi^4 = (1\ 2)(4\ 5\ 3)$

$\pi^5 = (2\ 1)(5\ 3\ 4)$

$\pi^6 = (1\ 2)(3\ 4\ 5) = \pi^0$

$\therefore$ Order $= 6.$

Theorem: The order of a finite set is L.C.M of the lengths of its disjoint cycles.

$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 2 & 4 \end{pmatrix}$, $o(f) = ?$

$f = (1\ 3\ 2)$

$\therefore o(f) = 3$

Note: If there is only a single cycle present, then order will be the length of the cycle.

$g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 4 & 2 & 3 & 1 & 7 & 8 & 6 \end{pmatrix}$, $o(g) = ?$

$g = (1\ 5)(2\ 4\ 3)(6\ 7\ 8)$

$\therefore o(f) = \text{LCM}(2, 3, 3)$

$= \underline{6}.$

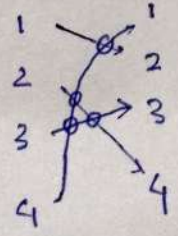
Even or odd permutation:

Product of even no. of transposition → Even permutation

Product of odd no. of ~~per~~ . → odd ..

$\beta = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix}$

Trick,



4 cross → ~~odd~~ even.

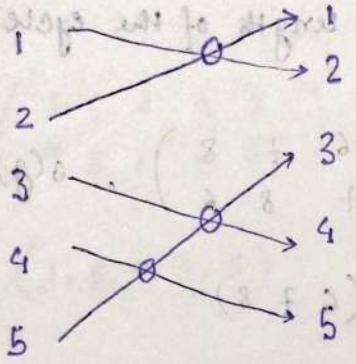
∴ even permutation.

$A = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix}$

$B = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{pmatrix}$

Parity (Even or odd) of AB ?

$AB = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 4 & 5 & 3 \end{pmatrix}$



3 cross → odd.

∴ odd parity.

Permutation group or symmetric group :

The group of all permutation of degree n, is called symmetric group.